

Lab 9

Question 1

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Tilix: thecoon@thecoon: ~  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ echo "All files in my home directory" > file_report.txt  
thecoon@thecoon:~$ ls -A ~ | tr '\n' ' ' >> file_report.txt  
thecoon@thecoon:~$ cat file_report.txt  
All files in my home directory  
access.log,AlphaProject,.bash_aliases,.bash_history,.bash_logout,.bashrc,.cache,cis106,.config,.copilot,Desktop,Documents,.dotnet,Downloads,earch --names-only tetris,essentials.sh,.face,.face  
.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hello.sh,inClassActivity,lab5files,lab5files.zip,.local,man-page-activity.pdf,marstenHouse,marstenhouse.sh,marstenhouse.sh.1,marstenhouse.sh.2,m  
arstenhouse.sh.3,marstenhouse.sh.4,.mozilla,Music,Pictures,.pki,practicecd,practicels,.profile,ProjectDelta,Public,scripts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.lo  
g,.ssh,.sudo_as_admin_successful,systemInfo.sh,Templates,TESTFOLDER,user_def.sh,UsersExport.csv,UsersExport.tsv,.vboxclient-clipboard-tty2-control.pid,.vboxclient-clipboard-tty2-service.pid,.  
vboxclient-display-svg-x11-tty2-control.pid,.vboxclient-display-tty2-control.pid,.vboxclient-display-tty63-control.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostversion-tty2-c  
ontrol.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmvga-session-tty2-control.pid,.vboxclient-vmvga-session-tty2-service.pid,Videos,.vscode,vscode.sh,.vscode-shared,website_ch1_la  
b7,.wget-hsts,wildcard_extra_practice,wildcardpractice,thecoon@thecoon:~$
```

Question 2

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Tilix: thecoon@thecoon: ~  
2: thecoon@thecoon: ~  
thecoon@thecoon:~$ ls -LA ~ | grep "^- " | wc -l >> file_report.txt  
thecoon@thecoon:~$ echo "Total number of directories in my home directory" >> file_report.txt  
thecoon@thecoon:~$ ls -LA ~ | grep "^d" | wc -l >> file_report.txt  
thecoon@thecoon:~$ tail -n 10 file_report.txt  
All files in my home directory  
access.log,AlphaProject,.bash_aliases,.bash_history,.bash_logout,.bashrc,.cache,cis106,.config,.copilot,Desktop,Documents,.dotnet,Downloads,earch --names-only  
tetris,essentials.sh,.face,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hello.sh,inClassActivity,lab5files,lab5files.zip,.local,man-page-activity.pdf,ma  
rstenHouse,marstenhouse.sh,marstenhouse.sh.1,marstenhouse.sh.2,marstenhouse.sh.3,marstenhouse.sh.4,.mozilla,Music,Pictures,.pki,practicecd,practicels,.profile,  
ProjectDelta,Public,scripts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.log,.ssh,.sudo_as_admin_successful,systemInfo.sh,Templates,TESTFO  
LDER,user_def.sh,UsersExport.csv,UsersExport.tsv,.vboxclient-clipboard-tty2-control.pid,.vboxclient-clipboard-tty2-service.pid,.vboxclient-display-svg-x11-tty  
2-control.pid,.vboxclient-display-tty2-control.pid,.vboxclient-display-tty63-control.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostversion-tty2-  
control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmvga-session-tty2-control.pid,.vboxclient-vmvga-session-tty2-service.pid,Videos,.vscode,vscode  
.sh,.vscode-shared,website_ch1_lab7,.wget-hsts,wildcard_extra_practice,wildcardpractice,Total number of files in my home directory  
0  
Total number of directories in my home directory  
34  
thecoon@thecoon:~$ cat file_report.txt  
All files in my home directory  
access.log,AlphaProject,.bash_aliases,.bash_history,.bash_logout,.bashrc,.cache,cis106,.config,.copilot,Desktop,Documents,.dotnet,Downloads,earch --names-only  
tetris,essentials.sh,.face,.face.icon,file_report.txt,.fonts,.gitconfig,.gnupg,hello.sh,inClassActivity,lab5files,lab5files.zip,.local,man-page-activity.pdf,ma  
rstenHouse,marstenhouse.sh,marstenhouse.sh.1,marstenhouse.sh.2,marstenhouse.sh.3,marstenhouse.sh.4,.mozilla,Music,Pictures,.pki,practicecd,practicels,.profile,  
ProjectDelta,Public,scripts,snap,sorted_access_log_by_endpoint.log,sorted_access_log_by_uname.log,.ssh,.sudo_as_admin_successful,systemInfo.sh,Templates,TESTFO  
LDER,user_def.sh,UsersExport.csv,UsersExport.tsv,.vboxclient-clipboard-tty2-control.pid,.vboxclient-clipboard-tty2-service.pid,.vboxclient-display-svg-x11-tty  
2-control.pid,.vboxclient-display-tty2-control.pid,.vboxclient-display-tty63-control.pid,.vboxclient-draganddrop-tty2-control.pid,.vboxclient-hostversion-tty2-  
control.pid,.vboxclient-seamless-tty2-control.pid,.vboxclient-vmvga-session-tty2-control.pid,.vboxclient-vmvga-session-tty2-service.pid,Videos,.vscode,vscode  
.sh,.vscode-shared,website_ch1_lab7,.wget-hsts,wildcard_extra_practice,wildcardpractice,Total number of files in my home directory  
0  
Total number of directories in my home directory  
34  
thecoon@thecoon:~$
```

Question 3

```
2/2 v + D Q Tilix: thecoon@thecoon: ~
1: thecoon@thecoon: ~
thecoon@thecoon:~$ awk -F: '{print $1}' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
sane
polkitd
rtkit
colord
...

2/2 v + D Q Tilix: thecoon@thecoon: ~
1: thecoon@thecoon: ~
thecoon@thecoon:~$ awk -F: '{print $1, $NF}' /etc/passwd
root /bin/bash
daemon /usr/sbin/nologin
bin /usr/sbin/nologin
sys /usr/sbin/nologin
sync /bin/sync
games /usr/sbin/nologin
man /usr/sbin/nologin
lp /usr/sbin/nologin
mail /usr/sbin/nologin
news /usr/sbin/nologin
uucp /usr/sbin/nologin
proxy /usr/sbin/nologin
www-data /usr/sbin/nologin
backup /usr/sbin/nologin
list /usr/sbin/nologin
irc /usr/sbin/nologin
_apt /usr/sbin/nologin
nobody /usr/sbin/nologin
systemd-network /usr/sbin/nologin
dhcpcd /bin/false
tss /bin/false
systemd-timesync /usr/sbin/nologin
messagebus /usr/sbin/nologin
dnsmasq /usr/sbin/nologin
avahi /usr/sbin/nologin
speech-dispatcher /bin/false
usbmux /usr/sbin/nologin
cups-pk-helper /usr/sbin/nologin
fwupd-refresh /usr/sbin/nologin
geoclue /usr/sbin/nologin
gnome-remote-desktop /usr/sbin/nologin
sane /usr/sbin/nologin
polkitd /usr/sbin/nologin
rtkit /usr/sbin/nologin
colord /usr/sbin/nologin
...

2/2 v + D Q Tilix: thecoon@thecoon: ~
1: thecoon@thecoon: ~
thecoon@thecoon:~$ awk -F: '{print $1, $7, $6}' /etc/passwd
root /bin/bash /root
daemon /usr/sbin/nologin /usr/sbin
bin /usr/sbin/nologin /bin
sys /usr/sbin/nologin /dev
sync /bin/sync /bin
games /usr/sbin/nologin /usr/games
man /usr/sbin/nologin /var/cache/man
lp /usr/sbin/nologin /var/spool/lpd
mail /usr/sbin/nologin /var/mail
news /usr/sbin/nologin /var/spool/news
uucp /usr/sbin/nologin /var/spool/uucp
proxy /usr/sbin/nologin /bin
www-data /usr/sbin/nologin /var/www
backup /usr/sbin/nologin /var/backups
list /usr/sbin/nologin /var/list
irc /usr/sbin/nologin /run/ircd
_apt /usr/sbin/nologin /nonexistent
nobody /usr/sbin/nologin /nonexistent
systemd-network /usr/sbin/nologin /
dhcpcd /bin/false /usr/lib/dhcpcd
tss /bin/false /var/lib/tpm
systemd-timesync /usr/sbin/nologin /
messagebus /usr/sbin/nologin /nonexistent
dnsmasq /usr/sbin/nologin /var/lib/misc
avahi /usr/sbin/nologin /run/avahi-daemon
speech-dispatcher /bin/false /run/speech-dispatcher
usbmux /usr/sbin/nologin /var/lib/usbmux
cups-pk-helper /usr/sbin/nologin /nonexistent
fwupd-refresh /usr/sbin/nologin /var/lib/fwupd
geoclue /usr/sbin/nologin /var/lib/geoclue
gnome-remote-desktop /usr/sbin/nologin /var/lib/gnome-remote-desktop
sane /usr/sbin/nologin /var/lib/sane
polkitd /usr/sbin/nologin /
rtkit /usr/sbin/nologin /proc
colord /usr/sbin/nologin /var/lib/colord
...
```

```

Tilix: thecoon@thecoon: ~
thecoon@thecoon:~$ awk -F: '{print "The user:", $1, "logs in with:", $7, "and is home dir is:", $6}' /etc/passwd
The user: root logs in with: /bin/bash and is home dir is: /root
The user: daemon logs in with: /usr/sbin/nologin and is home dir is: /usr/sbin
The user: bin logs in with: /usr/sbin/nologin and is home dir is: /bin
The user: sys logs in with: /usr/sbin/nologin and is home dir is: /dev
The user: sync logs in with: /bin/sync and is home dir is: /bin
The user: games logs in with: /usr/sbin/nologin and is home dir is: /usr/games
The user: man logs in with: /usr/sbin/nologin and is home dir is: /var/cache/man
The user: lp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/lpd
The user: mail logs in with: /usr/sbin/nologin and is home dir is: /var/mail
The user: news logs in with: /usr/sbin/nologin and is home dir is: /var/spool/news
The user: uucp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/uucp
The user: proxy logs in with: /usr/sbin/nologin and is home dir is: /bin
The user: www-data logs in with: /usr/sbin/nologin and is home dir is: /var/www
The user: backup logs in with: /usr/sbin/nologin and is home dir is: /var/backups
The user: list logs in with: /usr/sbin/nologin and is home dir is: /var/list
The user: irc logs in with: /usr/sbin/nologin and is home dir is: /run/ircd
The user: _apt logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: nobody logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: systemd-network logs in with: /usr/sbin/nologin and is home dir is: /
The user: dhcpcd logs in with: /bin/false and is home dir is: /usr/lib/dhcpcd
The user: tss logs in with: /bin/false and is home dir is: /var/lib/tpm
The user: systemd-timesync logs in with: /usr/sbin/nologin and is home dir is: /
The user: messagebus logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: dnsmasq logs in with: /usr/sbin/nologin and is home dir is: /var/lib/misc
The user: avahi logs in with: /usr/sbin/nologin and is home dir is: /run/avahi-daemon
The user: speech-dispatcher logs in with: /bin/false and is home dir is: /run/speech-dispatcher
The user: usbmux logs in with: /usr/sbin/nologin and is home dir is: /var/lib/usbmux
The user: cups-pk-helper logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: fwupd-refresh logs in with: /usr/sbin/nologin and is home dir is: /var/lib/fwupd
The user: geoclue logs in with: /usr/sbin/nologin and is home dir is: /var/lib/geoclue
The user: gnome-remote-desktop logs in with: /usr/sbin/nologin and is home dir is: /var/lib/gnome-remote-desktop
The user: saned logs in with: /usr/sbin/nologin and is home dir is: /var/lib/saned
The user: polkitd logs in with: /usr/sbin/nologin and is home dir is: /
The user: rtkit logs in with: /usr/sbin/nologin and is home dir is: /proc
The user: colord logs in with: /usr/sbin/nologin and is home dir is: /var/lib/colord

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Tilix: thecoon@thecoon: ~
thecoon@thecoon:~$ awk -F: '{print "The user:", $1, "logs in with:", $7, "and is home dir is:", $6}' /etc/passwd > report_users.txt
thecoon@thecoon:~$ grep "/bin/bash" /etc/passwd
root:x:0:0:root:/root:/bin/bash
thecoon:x:1000:1000:thecoon,,/home/thecoon:/bin/bash
thecoon@thecoon:~$ grep "/bin/bash" /etc/passwd | awk -F: '{print $1 " = " $7}'
root = /bin/bash
thecoon = /bin/bash
thecoon@thecoon:~$ grep "/bin/bash" /etc/passwd | awk -F: '{print $1 " = " $7}' >> report_users.txt
thecoon@thecoon:~$

```

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Tilix: thecoon@thecoon: ~
thecoon@thecoon:~$ cat report_users.txt
The user: root logs in with: /bin/bash and is home dir is: /root
The user: daemon logs in with: /usr/sbin/nologin and is home dir is: /usr/sbin
The user: bin logs in with: /usr/sbin/nologin and is home dir is: /bin
The user: sys logs in with: /usr/sbin/nologin and is home dir is: /dev
The user: sync logs in with: /bin/sync and is home dir is: /bin
The user: games logs in with: /usr/sbin/nologin and is home dir is: /usr/games
The user: man logs in with: /usr/sbin/nologin and is home dir is: /var/cache/man
The user: lp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/lpd
The user: mail logs in with: /usr/sbin/nologin and is home dir is: /var/mail
The user: news logs in with: /usr/sbin/nologin and is home dir is: /var/spool/news
The user: uucp logs in with: /usr/sbin/nologin and is home dir is: /var/spool/uucp
The user: proxy logs in with: /usr/sbin/nologin and is home dir is: /bin
The user: www-data logs in with: /usr/sbin/nologin and is home dir is: /var/www
The user: backup logs in with: /usr/sbin/nologin and is home dir is: /var/backups
The user: list logs in with: /usr/sbin/nologin and is home dir is: /var/list
The user: irc logs in with: /usr/sbin/nologin and is home dir is: /run/ircd
The user: _apt logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: nobody logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: systemd-network logs in with: /usr/sbin/nologin and is home dir is: /
The user: dhcpcd logs in with: /bin/false and is home dir is: /usr/lib/dhcpcd
The user: tss logs in with: /bin/false and is home dir is: /var/lib/tpm
The user: systemd-timesync logs in with: /usr/sbin/nologin and is home dir is: /
The user: messagebus logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: dnsmasq logs in with: /usr/sbin/nologin and is home dir is: /var/lib/misc
The user: avahi logs in with: /usr/sbin/nologin and is home dir is: /run/avahi-daemon
The user: speech-dispatcher logs in with: /bin/false and is home dir is: /run/speech-dispatcher
The user: usbmux logs in with: /usr/sbin/nologin and is home dir is: /var/lib/usbmux
The user: cups-pk-helper logs in with: /usr/sbin/nologin and is home dir is: /nonexistent
The user: fwupd-refresh logs in with: /usr/sbin/nologin and is home dir is: /var/lib/fwupd
The user: geoclue logs in with: /usr/sbin/nologin and is home dir is: /var/lib/geoclue
The user: gnome-remote-desktop logs in with: /usr/sbin/nologin and is home dir is: /var/lib/gnome-remote-desktop
The user: saned logs in with: /usr/sbin/nologin and is home dir is: /var/lib/saned
The user: polkitd logs in with: /usr/sbin/nologin and is home dir is: /
The user: rtkit logs in with: /usr/sbin/nologin and is home dir is: /proc
The user: colord logs in with: /usr/sbin/nologin and is home dir is: /var/lib/colord

```

Question 4

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1: thecoon@thecoon: ~
thecoon@thecoon:~$ head -n 1 ~/Documents/sample_files/Csv/cars.csv > honda1.csv
thecoon@thecoon:~$ grep "Honda" ~/Documents/sample_files/csv/cars.csv >> honda1.csv
thecoon@thecoon:~$ sed 's/;/,/g' honda1.csv > honda2.csv
thecoon@thecoon:~$ echo "model,cylinders,hp" > Honda3.csv
thecoon@thecoon:~$ awk -F, 'NR>1 {print $1 "," $3 "," $5}' honda2.csv
Honda Civic,4,97.00
Honda Civic CVCC,4,53.00
Honda Civic,4,53.00
Honda Accord CVCC,4,68.00
Honda Civic CVCC,4,60.00
Honda Accord LX,4,68.00
Honda Civic 1500 gl,4,67.00
Honda Accord,4,72.00
Honda Civic 1300,4,60.00
Honda Prelude,4,75.00
Honda Accord,4,75.00
Honda Civic,4,67.00
Honda Civic (auto),4,67.00
thecoon@thecoon:~$ awk -F, 'NR>1 {print $1 "," $3 "," $5}' honda2.csv \
> | sed 's/,/,/g' \
> | sed 's/Honda /Honda_/g' \
> >> Honda3.csv
thecoon@thecoon:~$

2: thecoon@thecoon: ~
thecoon@thecoon:~$ cat honda1.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
thecoon@thecoon:~$ cat honda1.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
Honda Civic;24.0;4;120.0;97.00;2489.;15.0;74;Japan
Honda Civic CVCC;33.0;4;91.00;53.00;1795.;17.5;75;Japan
Honda Civic;33.0;4;91.00;53.00;1795.;17.4;76;Japan
Honda Accord CVCC;31.5;4;98.00;68.00;2045.;18.5;77;Japan
Honda Civic CVCC;36.1;4;91.00;60.00;1800.;16.4;78;Japan
Honda Accord LX;29.5;4;98.00;68.00;2135.;16.6;78;Japan
Honda Civic 1500 gl;44.6;4;91.00;67.00;1850.;13.8;80;Japan
Honda Accord;32.4;4;107.0;72.00;2290.;17.0;80;Japan
Honda Civic 1300;35.1;4;81.00;60.00;1760.;16.1;81;Japan
Honda Prelude;33.7;4;107.0;75.00;2210.;14.4;81;Japan
Honda Accord;36.0;4;107.0;75.00;2205.;14.5;82;Japan
Honda Civic;38.0;4;91.00;67.00;1965.;15.0;82;Japan
Honda Civic (auto);32.0;4;91.00;67.00;1965.;15.7;82;Japan
thecoon@thecoon:~$

1: thecoon@thecoon: ~
thecoon@thecoon:~$ head -n 1 ~/Documents/sample_files/Csv/cars.csv > honda1.csv
thecoon@thecoon:~$ grep "Honda" ~/Documents/sample_files/csv/cars.csv >> honda1.csv
thecoon@thecoon:~$ sed 's/;/,/g' honda1.csv > honda2.csv
thecoon@thecoon:~$ echo "model,cylinders,hp" > Honda3.csv
thecoon@thecoon:~$ awk -F, 'NR>1 {print $1 "," $3 "," $5}' honda2.csv
Honda Civic,4,97.00
Honda Civic CVCC,4,53.00
Honda Civic,4,53.00
Honda Accord CVCC,4,68.00
Honda Civic CVCC,4,60.00
Honda Accord LX,4,68.00
Honda Civic 1500 gl,4,67.00
Honda Accord,4,72.00
Honda Civic 1300,4,60.00
Honda Prelude,4,75.00
Honda Accord,4,75.00
Honda Civic,4,67.00
Honda Civic (auto),4,67.00
thecoon@thecoon:~$ awk -F, 'NR>1 {print $1 "," $3 "," $5}' honda2.csv | sed 's/,/,/g'
Honda Civic,4,97.00
Honda Civic CVCC,4,53.00
Honda Civic,4,53.00
Honda Accord CVCC,4,68.00
Honda Civic CVCC,4,60.00
Honda Accord LX,4,68.00
Honda Civic 1500 gl,4,67.00
Honda Accord,4,72.00
Honda Civic 1300,4,60.00
Honda Prelude,4,75.00
Honda Accord,4,75.00
Honda Civic,4,67.00
Honda Civic (auto),4,67.00
thecoon@thecoon:~$ awk -F, 'NR>1 {print $1 "," $3 "," $5}' honda2.csv | sed 's/Honda /Honda_/g'
>> Honda3.csv
thecoon@thecoon:~$

2: thecoon@thecoon: ~
thecoon@thecoon:~$ batcat Honda3.csv
File: Honda3.csv
1 model,cylinders,hp
2 Honda_Civic,4,97.00
3 Honda_Civic CVCC,4,53.00
4 Honda_Civic,4,53.00
5 Honda_Accord CVCC,4,68.00
6 Honda_Civic CVCC,4,60.00
7 Honda_Accord LX,4,68.00
8 Honda_Civic 1500 gl,4,67.00
9 Honda_Accord,4,72.00
10 Honda_Civic 1300,4,60.00
11 Honda_Prelude,4,75.00
12 Honda_Accord,4,75.00
13 Honda_Civic,4,67.00
14 Honda_Civic (auto),4,67.00
15 Honda_Civic,4,97.00
16 Honda_Civic CVCC,4,53.00
17 Honda_Civic,4,53.00
18 Honda_Accord CVCC,4,68.00
19 Honda_Civic CVCC,4,60.00
20 Honda_Accord LX,4,68.00
21 Honda_Civic 1500 gl,4,67.00
22 Honda_Accord,4,72.00
23 Honda_Civic 1300,4,60.00
24 Honda_Prelude,4,75.00
25 Honda_Accord,4,75.00
26 Honda_Civic,4,67.00
27 Honda_Civic (auto),4,67.00
thecoon@thecoon:~$

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